

Experiment 2. Case study on use of Simulink in MATLAB for engineering problems

Aim: To analyse the step response of a first- order system using MATLAB SIMULINK

A typical first- order transfer function is:

$$G(s)=K/\tau s+1$$

Where:

K = Gain

τ = Time constant

Procedure:

1: Open MATLAB, Start MATLAB

- In command window, type: **simulink**

2: Create New Model

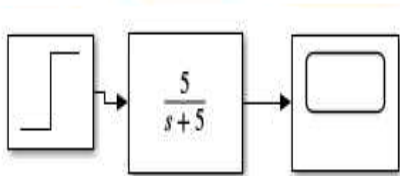
- Click “**Blank Model**”
- Save file as: first_order_system.slx

3: Insert Required Blocks

From Simulink Library Browser:

- Step (Sources)
- Transfer Function (Continuous)
- Scope (Sinks)

4: Connect Blocks



Step → Transfer Function → Scope

5: Define Parameters

Step Block:

- Step time: 0
- Initial value: 0
- Final value: 5

Transfer Function Block:

- Numerator: [5]
- Denominator: [1 5]
- (Represents $1/2s+1$)

6: Simulation Settings

- Click Simulation → Model Settings
- Stop time: 10

7: Run Simulation

- Click Run (button)

8: Analyze Output

- Double-click Scope (You will see the step response curve)

9.Expected Output

- Exponential rise
- Starts from 0
- Settles at 1 (steady state value)

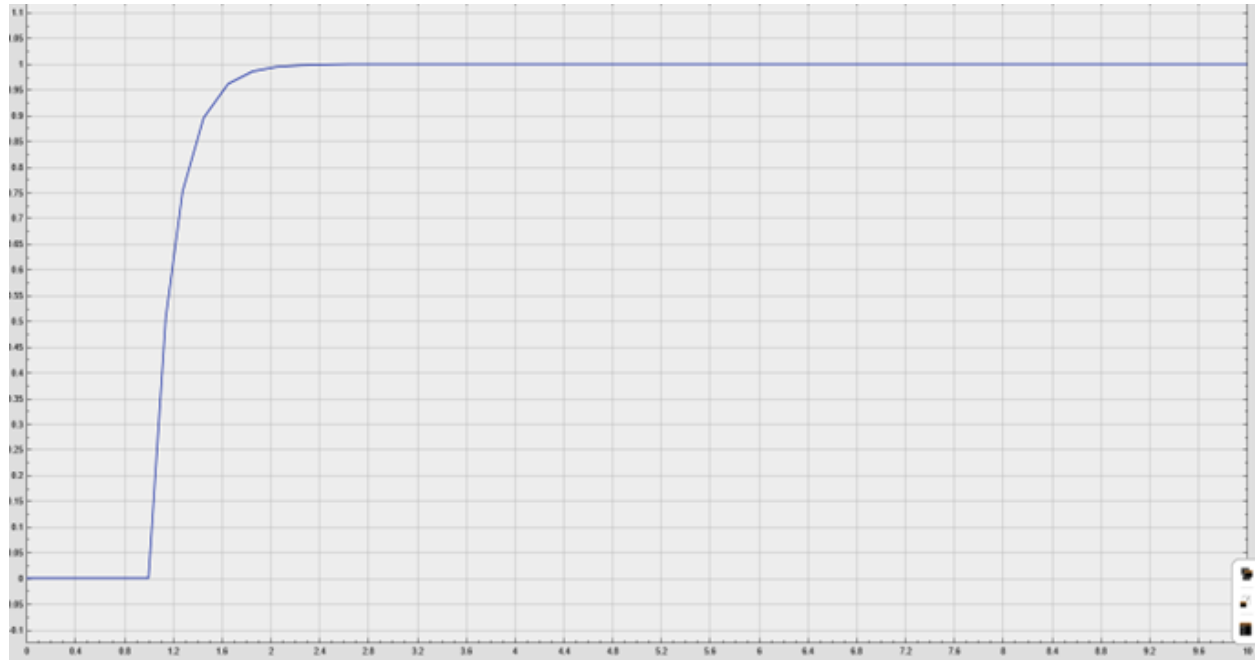
Time constant $\tau = 2$

System reaches ~63% of final value at

$t = 2$

Settling time $\approx 4\tau = 8$

Key observations:



Conclusion

The step response of a first-order system was successfully analyzed using MATLAB Simulink. The simulation results show that the system responds smoothly to a step input without oscillations. The output gradually reaches the steady-state value, demonstrating the characteristic behavior of a first-order system.